S3.2 Functional groups: Classification of organic compounds

Stereoisomers: cis-trans isomerism (HL)

Higher level (HL)

Learning outcomes

At the end of this section you should be able to describe and explain the features that give rise to *cis-trans* isomerism; recognise it in non-cyclic alkenes and C3 and C4 cycloalkanes.

How many different isomers do you think there are with the molecular formula $C_{10}H_{22}$? The answer is 136, which does not seem that surprising considering carbon's ability to form chains and rings with itself. How about with the molecular $C_{20}H_{42}$? More than 3 000 000! And finally, how many different isomers do you think there are there with the molecular formula $C_{40}H_{82}$? Amazingly, there are more than 13 000 000 000 000 000 (thirteen quadrillion) different isomers that can be made with 40 carbon atoms and 82 hydrogen atoms. Luckily, you do not have to try and name them all.

In this section, you will learn about a form of isomerism known as stereoisomerism. Stereoisomers are compounds that have the same structural formulas but a different spatial arrangement of atoms. In other words, they have the same bond connectivities but the atoms have different orientations in space. Stereoisomers can be further divided into *cis-trans* isomers and optical isomers (**Figure 1**). Other types of stereoisomerism exist but are not covered in the DP chemistry course. In this section, we will look at *cis-trans* isomers and in the next section 3.2.7b (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/stereoisomers-optical-isomerism-hl-id-45667/), optical isomers.

Figure 1. Stereoisomers can be divided into *cis-trans* isomers and optical isomers.

More information for figure 1

The image is a diagram illustrating the division of stereoisomers into two categories: cis-trans isomers and optical isomers. At the top, the word "Stereoisomers" is enclosed in a rectangular box, with a line extending downwards to a horizontal line. This horizontal line branches into two separate vertical lines, each leading to another rectangular box. The box on the left contains the label "cis-trans isomers," while the box on the right contains "Optical isomers." This structure visually represents the categorization of stereoisomers into these two types, showing a simple hierarchical relationship.

[Generated by AI]

Cis-trans isomerism occurs in molecules that have restricted rotation somewhere in the molecule, which can be in the form of a C=C bond or a ring structure. To help you understand the concept of restricted rotation, try this activity.

1. Take a pen and push it through two sheets of paper as shown in **Figure 2**. Try to rotate each piece of paper.

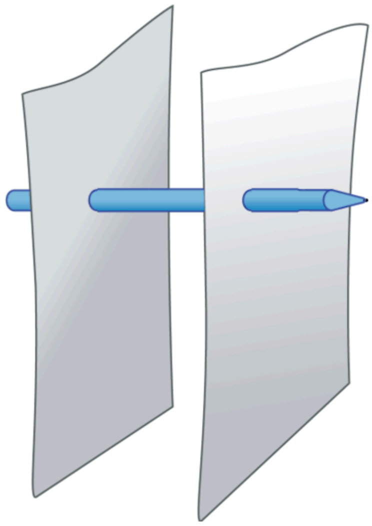


Figure 2. Push a pen through two sheets of paper.

1. Next, take a second pen and push it through both pieces of paper as shown in **Figure 3**. Now try to rotate the pieces of paper. What do you notice?

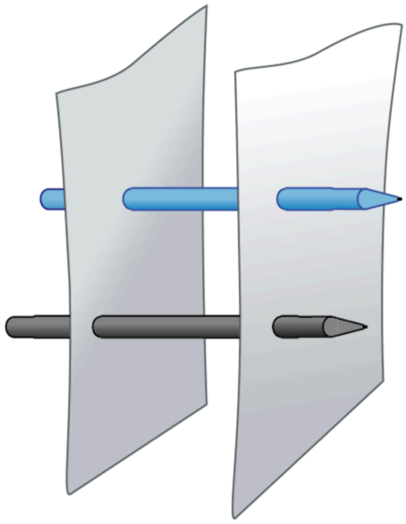


Figure 3. Now push a second pen through the same two sheets of paper.

You may have noticed that it was possible to rotate the pieces of paper with one pen but not two. The single pen represents a single covalent bond that is made up of a sigma bond that allows free rotation. The addition of the second pen represents a double covalent bond that is composed of one sigma and one pi bond. The pi bond prevents free rotation around the C=C bond in the molecule. There is also restricted rotation in cyclic compounds that have single C–C bonds because the bond rotation is prevented by the ring structure itself (F**igure 4**).

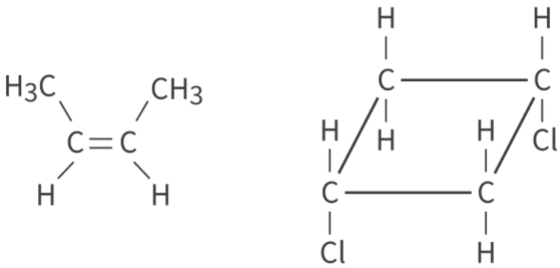


Figure 4. Compounds with C=C bonds (left) or a ring structure (right) have restricted rotation of the bonds.

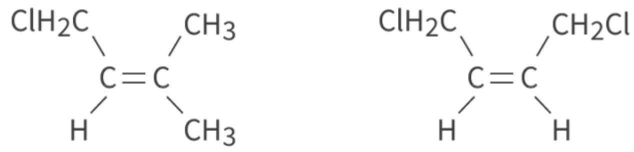
More information for figure 4

The image displays two chemical structures. On the left is a molecule with a double bond, specifically a C=C bond, where each carbon atom is bonded to a methyl group (CH3) and a hydrogen atom (H). The presence of the C=C double bond restricts the rotation of the bonds within the molecule. On the right is a cyclic compound, a ring structure composed of carbon and hydrogen atoms, with chlorines (Cl) substituents on two of the carbon atoms. This ring structure also restricts rotation due to its cyclic nature.

[Generated by AI]

This restricted rotation results in two isomers that differ in the spatial arrangement of the atoms. The two isomers can only be interconverted by the breaking of the bonds between the atoms. The first type of isomer we will look at is compounds that have restricted rotation because they have a C=C bond.

For *cis-trans* isomerism to occur, there must be two different atoms, or groups of atoms, bonded to each of the carbon atoms of the C=C bond. Take the two compounds shown in **Figure 5** for example. 1-chloro-3-methylbut-2-ene (left) only has two different groups bonded to one of the carbon atoms of the C=C bond so does not have *cis-trans* isomers. 1,4-dichlorobut-2-ene (right) has two different groups bonded to each of the carbon atoms of the C=C bond so it does have *cis-trans* isomers.



1-chloro-3-methylbut-2-ene   1,4-dichlorobut-2-ene

Figure 5. 1-chloro-3-methylbut-2-ene cannot exist as *cis-trans* isomers but 1,4-dichlorobut-2-ene can. Can you explain why?

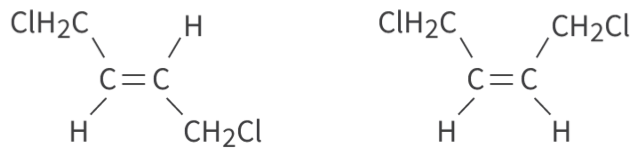
More information for figure 5.1

The image shows the structural formulas of two chemical compounds. On the left is 1-chloro-3-methylbut-2-ene, which lacks the capacity for cis-trans isomerism. Its carbon-carbon double bond (C=C) connects two carbon atoms, with a chloromethyl group (CH2Cl) and a hydrogen atom (H) bonded to one carbon, and two methyl groups (CH3) bonded to the other carbon, which provides no variation at one carbon for cis-trans isomerism.

On the right is 1,4-dichlorobut-2-ene, which can exist as cis-trans isomers. Its C=C bond has a chloromethyl group (CH2Cl) and hydrogen atom (H) bonded to each carbon. This configuration allows for the existence of cis forms, where the CH2Cl groups are on the same side, and trans forms, where they are on opposite sides of the double bond.

[Generated by AI]

The restricted rotation of the C=C bond results in two isomers. The *trans*-isomer has the same groups on opposite sides of the C=C bond and the *cis*-isomer has the same groups on the same side. The compounds are named *trans*-1,4-dichlorobut-2-ene and *cis*-1,4-dichlorobut-2-ene, respectively (**Figure 6**).



trans-1,4-dichlorobut-2-ene   *cis*-1,4-dichlorobut-2-ene

Figure 6. The full structural formulas of *cis* and *trans* 1,4-dichlorobut-2-ene.

More information for figure 6

The image shows the structural formulas for trans-1,4-dichlorobut-2-ene and cis-1,4-dichlorobut-2-ene. On the left, the trans-isomer displays chlorine and hydrogen atoms on opposite sides of the central carbon-carbon double bond (C=C). The chlorine atom (CH2Cl) is attached to the first carbon, and the hydrogen atom (H) is attached to the second carbon, with the same configuration on the opposite side. On the right, the cis-isomer shows the same groups on the same side of the C=C bond. In this case, both chlorine atoms are on one side of the double bond, and both hydrogen atoms are on the opposite side. This demonstrates the restricted rotation around the C=C bond causing the formation of these two isomers.

[Generated by AI]

Worked example 1

The two *cis-trans* isomers of but-2-ene are shown below. Classify each isomer as the *cis*-isomer or the *trans*-isomer.

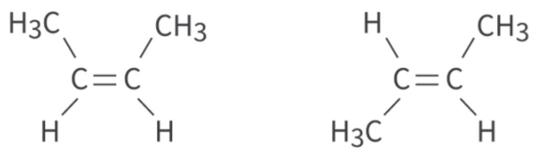


Figure 7. Classify the *cis*-isomer and the *trans*-isomer.

More information for figure 7

The image displays two structural formulas of but-2-ene, illustrating its cis and trans isomers. On the left is the cis-isomer, where both methyl groups (CH3) are on the same side of the double bond. The structure is represented as CH3−CH=CH−CH3, with both CH3 groups aligned vertically. On the right is the trans-isomer, where the methyl groups are on opposite sides of the double bond. It is represented as H−C=CH−CH3 with CH3 on the top and bottom of the structure, indicating opposite positioning.

[Generated by AI]

On the left we have *cis*-but-2-ene with both $—CH_3$ groups on the same side of the double bond. On the right is *trans*-but-2-ene with both $—CH_3$ groups on opposite sides of the double bond.

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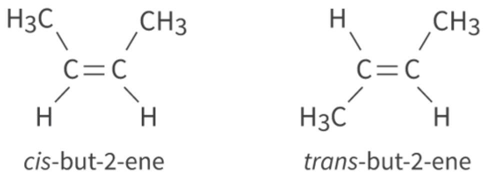

Next, we will look at isomers that have restricted rotation due to them having a ring structure. **Figure 8** shows two isomers that have ring structures. Both compounds have two chlorine atoms bonded to the carbon atoms numbered 1 and 3. The compound on the left has both chlorine atoms orientated below the plane of the ring. This is the *cis*-isomer. The molecule on the right has one chlorine atom below the plane and one chlorine atom above the plane of the ring. This is the *trans*-isomer.

The isomers are named *cis*-1,3-dichlorocyclobutane and *trans*-1,3-dichlorocyclobutane, respectively.

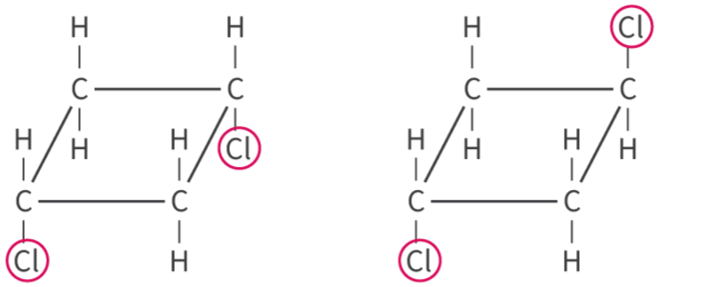

Figure 8. The *cis-trans* isomers of 1,3-dichlorocyclobutane.

More information for figure 8

The image depicts the chemical structure diagrams for the cis and trans isomers of 1,3-dichlorocyclobutane. On the left is the cis-isomer, where both chlorine atoms (Cl) are depicted below the plane of the cyclobutane ring. The cyclobutane ring consists of four carbon (C) atoms bonded in a square formation, with hydrogen (H) atoms filling the remaining valences. Each chlorine atom is bonded to the first and third carbon atoms respectively.

On the right is the trans-isomer, where one chlorine atom is above the plane of the ring and the other is below. Similar to the cis-isomer, the cyclical structure consists of carbon atoms in a square, with hydrogens occupying the remaining bonding positions. The only difference is the spatial orientation of

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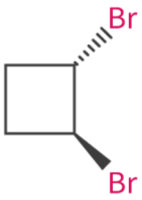
the chlorine atoms, which differentiates the trans configuration from the cis. The text under each structure provides additional information about the spatial orientation of chlorine in relation to the carbon ring plane.

[Generated by AI]

The *cis-trans* isomers with ring structures can also be represented using stereochemical formulas. **Figure 9** shows the stereochemical formula of *cis* and *trans* 1,2-dibromocyclobutane. Note that the *cis*-isomer has both bromine atoms above the plane of the ring and the *trans*-isomer has one bromine atom above and one bromine atom below the plane of the ring.

cis-1,2-dibromocyclobutane



trans-1,2-dibromocyclobutane

Figure 9. The *cis-trans* isomers of 1,2-dibromocyclobutane.

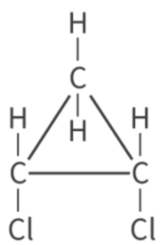
More information for figure 9

The image shows two stereochemical formulas of 1,2-dibromocyclobutane. On the left, the cis-1,2-dibromocyclobutane isomer is depicted with both bromine (Br) atoms above the plane of the cyclobutane ring. This is represented by solid wedges connecting each Br atom to adjacent carbon atoms of the square ring. On the right, the trans-1,2-dibromocyclobutane isomer is shown with one bromine atom above the plane (indicated with a solid wedge) and the other bromine atom below the plane (illustrated using dashed lines). The structural differences in bromine placement between the isomers are what distinguish the cis from the trans configuration.

[Generated by AI]

Worked example 2

Classify each compound shown below as either a *cis*-isomer or a *trans*-isomer.



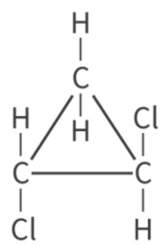


Figure 10. Classify the *cis*-isomer and the *trans*-isomer.

More information for figure 10

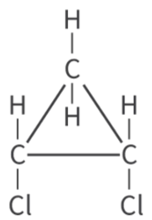
This image features two structural diagrams of a molecule, illustrating the cis-isomer and trans-isomer forms. Each diagram shows a three-carbon ring with chlorine (Cl) and hydrogen (H) atoms attached.

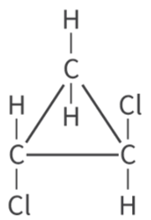
On the left, the diagram represents the cis-isomer. In this configuration, two chlorine atoms are on the same side of the ring. The carbon atoms at the vertices of the ring are each bonded to a hydrogen atom and a chlorine atom, maintaining the same orientation.

On the right, the diagram shows the trans-isomer. In this isomer, the chlorine atoms are on opposite sides of the ring, alternately bonded with hydrogen atoms on the opposite vertices, giving it a more staggered appearance. This contrast in orientation between the chlorine atoms differentiates the cis and trans forms of the molecule.

[Generated by AI]

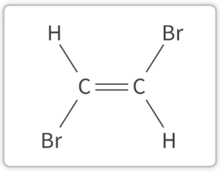
On the left is *cis*-1,2-dichlorocyclopropane which is the cis isomer. On the right is *trans*-1,2-dichlorocyclopropane which is the *trans*-isomer. The *cis*-isomer has both chlorine atoms below the pane of the ring and the *trans*-isomer has one chlorine atom below the plane of the ring and one above the plane of the ring.

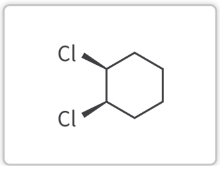

cis-1,2-dichlorocyclopropane

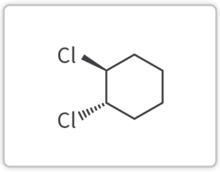

trans-1,2-dichlorocyclopropane

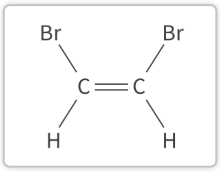
In **Interactive 1** you will practise matching pairs of *cis-trans* isomers.

***Cis*-isomer** ***Trans*-isomer**









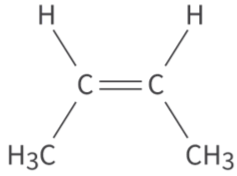
Check

Interactive 1. Practise matching pairs of *cis*-*trans* isomers.

Activity

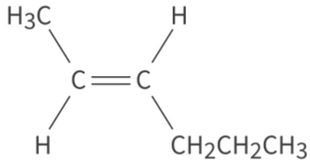
- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual activity

1. For each compound, state whether it is the *cis*- or *trans*-isomer.
   (a)



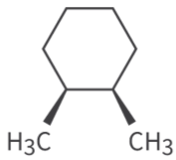
More information

1. (b)



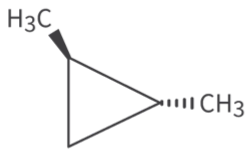
More information

(c)



More information

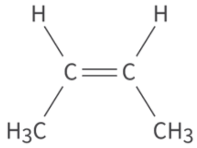
(d)



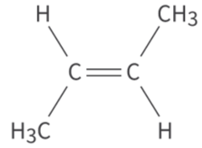
More information

2. For each isomer in part (1), sketch the *cis*- or *trans*- isomer for that compound (so if you identified the compound as the *cis*-isomer, sketch the *trans*-isomer and vice-versa).

1. (a) *cis*-isomer
   (b) *trans*-isomer
   (c) *cis*-isomer
   (d) *trans*-isomer
2. (a)

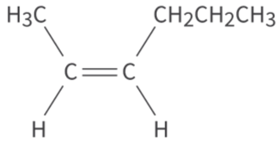


cis-isomer

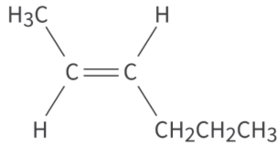


trans-isomer

(b)

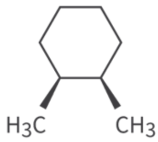


cis-isomer

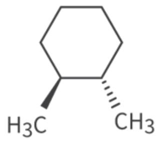


trans-isomer

(c)

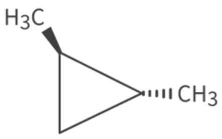


cis-isomer

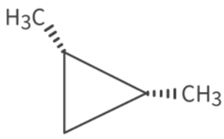


trans-isomer

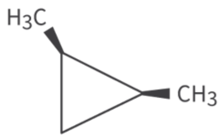
(d)



trans-isomer



or



cis-isomer

S3.2 Functional groups: Classification of organic compounds

Stereoisomers: optical isomerism (HL)

Higher level (HL)

Learning outcomes

At the end of this section you should be able to:

- Draw stereochemical formulas showing the tetrahedral arrangement around a chiral carbon.
- Describe and explain a chiral carbon atom giving rise to stereoisomers with different optical properties. Recognise a pair of enantiomers as non-superimposable mirror images from 3D modelling (real or virtual).

Why is it that a pair of gloves has a right-hand glove and a left-hand glove? Have you ever tried wearing a right-hand glove on your left hand and vice versa? Chances are, if you have tried, you realised that it is not possible. A right-hand glove only fits on your right hand and a left-hand glove only fits on your left hand. What is it about our hands that requires a different right- and left-hand glove? **Figure 1** shows what you would see if you placed your hand next to a mirror. What do you notice about the image?

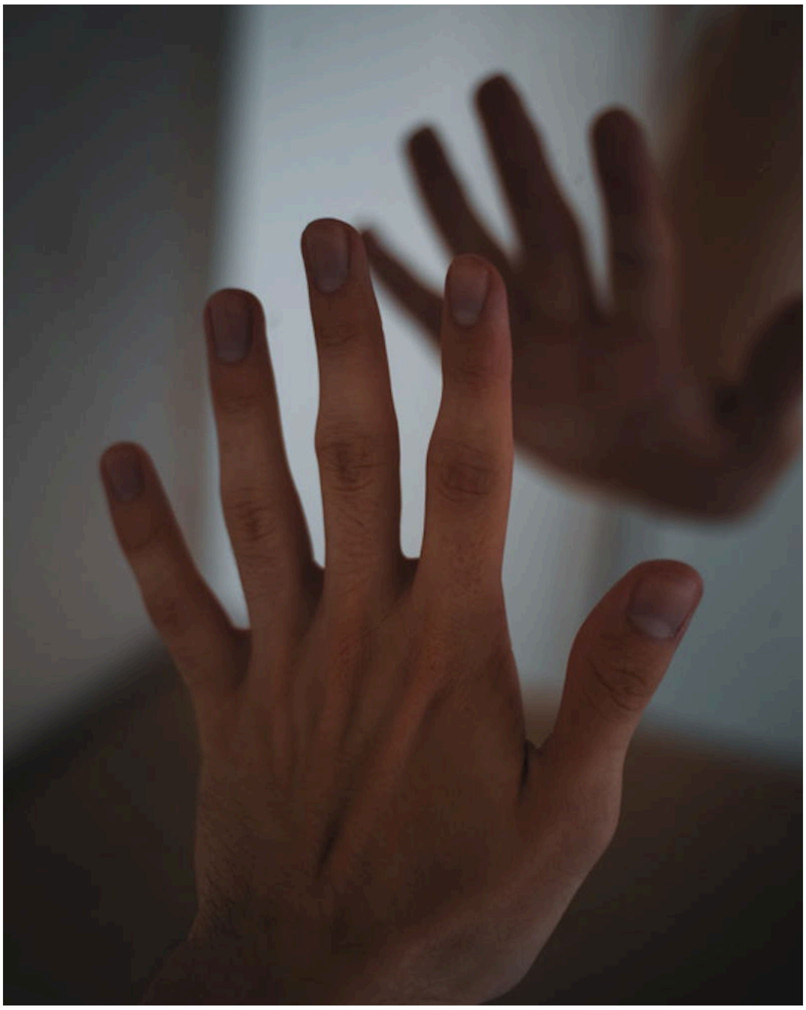
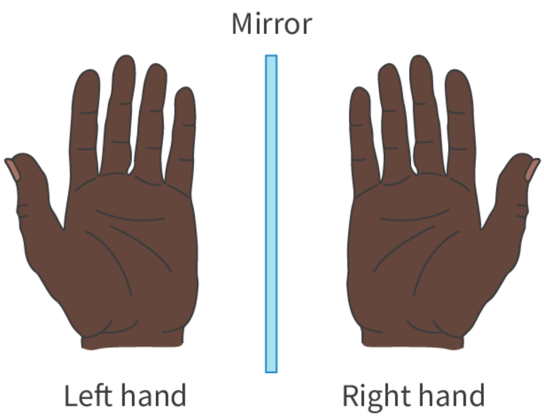


Figure 1. What do you notice about the reflection of the hand in the mirror?

Credit: Wirestock, Getty Images

Our hands are mirror images of each other; this is why we need separate right- and left-handed gloves. Because they are mirror images, they are non-superimposable, which means that they cannot be superimposed. Click on **Interactives 1 and 2** below to see the difference between a superimposable object and a non-superimposable object.



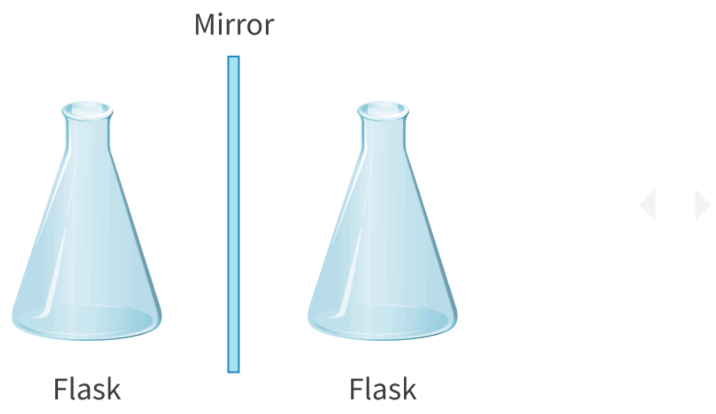
Interactive 1. Non-superimposable.

More information for interactive 1

This interactive exercise visually demonstrates the concept of non-superimposability using a familiar example: the left and right human hands. The activity is designed to help learners understand chirality. Chirality is a property of an object (or molecule) that means it cannot be superimposed on its mirror image.

A person's left hand is shown on the left, and their right hand is on the right side, with a mirror positioned between them. Users can move a slider across the image to reveal how the left and right hands align when overlaid. As the hands merge, it becomes clear that they do not match exactly, demonstrating that the hand is a non-superimposable object. The left hand's shape and structure do not match the right hand's, illustrating the concept of chirality, which also applies to many molecules.

This interactive exercise helps users to understand the concept of non-superimposability.



Interactive 2. Superimposable.

More information for interactive 2

An interactive image displays an example of a superimposable object. On the left side, a conical flask is placed next to a mirror, with its mirror image shown on the right.

Users can move a slider across the image to reveal how the original flask and its mirror image align when overlaid. As the two images merge, it becomes evident that the flask is a superimposable object because its shape and structure match its mirror image without any distinction.

This interactivity helps users to understand the concept of superimposability.

Look at **Figure 2** showing two earrings made with different coloured beads. Would you consider these two earrings to be the same? A quick look shows that they are the same size and they each contain the same number and colour of beads. Even though they have the same components, the two earrings are not truly identical as they are mirror images of each other. To make the beads have the same sequence left to right, one earring would have to be worn backwards. Think about other items you have encountered that might appear to be similar, but not identical.

Figure 2. The same components can have different orientations.

Like our hands, there are certain organic compounds that have the same molecular formula but are non-superimposable. The hypothetical compounds A and B shown in **Figure 3** have the same molecular formulas but they have different spatial arrangements of their atoms. What do you notice about these two molecules?

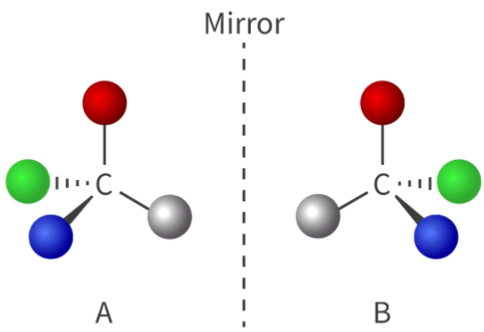


Figure 3. The two hypothetical compounds A and B.

More information for figure 3

The image shows two molecular structures labeled as A and B, positioned on either side of a vertical dashed line labeled 'Mirror.' Both compounds have the same molecular formula but are arranged in mirror image configurations. Each molecule consists of a central carbon atom, with four different colored atoms or groups attached, indicating a tetrahedral arrangement. The atoms are colored red, green, blue, and a light color (possibly gray or silver) representing different elements or groups. The arrangement of these groups around the central carbon is different in each molecule, illustrating a characteristic of chirality where the compounds are non-superimposable mirror images of each other, like human hands.

[Generated by AI]

You may have noticed that A and B are mirror images of each other, just like your hands. This property is known as chirality, named after the Greek word for hand. But what is it that gives molecules such as A and B this property? To explain this, let's consider a real chiral molecule such as butan-2-ol. The atoms in butan-2-ol can be arranged in two different ways that make the two forms non-superimposable. **Figure 4** shows these two forms, which are known as stereoisomers, optical isomers or enantiomers.

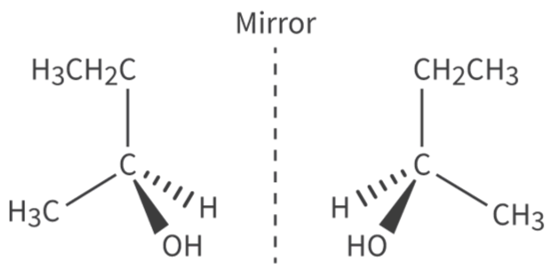


Figure 4. The two enantiomers of butan-2-ol.

More information for figure 4

The image is a diagram displaying the two enantiomers of butan-2-ol as mirror images. On the left side, the structure is labeled with ethyl group (H3C-CH2-C), hydroxyl group (OH), and hydrogen (H). The central carbon atom is chiral, creating a three-dimensional structure that includes bonds shown as dashed and solid wedges. The right side mirrors this arrangement, showing the ethyl group (CH2CH3) on the opposite side, with the hydroxyl and hydrogen group orientations reversed. A vertical dotted line labeled 'Mirror' separates the two structures, emphasizing their non-superimposable nature. This clearly demonstrates the concept of chirality, where the molecules are mirror images but cannot be superimposed on one another, illustrating the uniqueness of stereoisomers or optical isomers.

[Generated by AI]

For a molecule to have enantiomers, it must have a chiral carbon atom (also known as an asymmetric carbon atom or a chiral centre) within the molecule. A chiral carbon is a carbon atom that is bonded to four different atoms or groups. Compounds that contain a chiral carbon atom are said to be optically active and those that do not are optically inactive. This property can be used to distinguish between two enantiomers.

Kog

If we take another look at butan-2-ol (**Figure 5**), we can see that one carbon atom, marked with a red asterisk, is bonded to four different atoms or groups of atoms. The chiral carbon atom is bonded to an OH group, a CH_3 group, a C_2H_5 group and a hydrogen atom (note that the different lengths of the alkyl chains count as different groups). This makes a total of four different atoms or groups of atoms, therefore butan-2-ol is optically active.

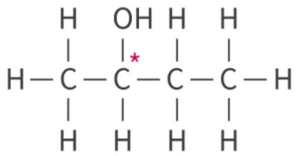


Figure 5. The full structural formula of butan-2-ol. The red asterisk indicates the chiral carbon atom.

More information for figure 5

The image shows the structural formula of butan-2-ol. It consists of a chain of four carbon atoms. From left to right, the first carbon atom is bonded to three hydrogen atoms. The second carbon atom is bonded to an OH group at the top, a hydrogen atom at the bottom, and a CH3 group on the left. This carbon is indicated as chiral with a red asterisk. The third carbon is bonded to two hydrogen atoms. The fourth carbon, at the right end, is bonded to three hydrogen atoms. The structural formula highlights that the chiral carbon is linked to four different groups: OH, CH3, C2H5 (from the chain), and H.

[Generated by AI]

Figure 6 shows the structure of the halogenoalkane 2-chlorobutane. Try to identify the chiral carbon atom in the molecule. When you are ready, check your answer below.

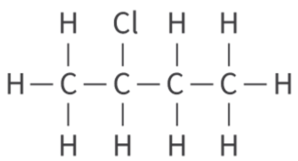


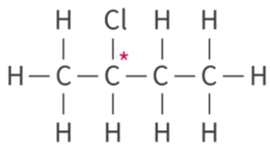
Figure 6. Try to identify the chiral carbon atom in 2-chlorobutane.

More information for figure 6

The image is a structural diagram of the organic compound 2-chlorobutane, which is a halogenoalkane. It illustrates a carbon chain with four carbon atoms labeled as C. Each carbon atom is bonded with hydrogen atoms (H) and there is one chlorine atom (Cl) substituting for a hydrogen atom around the second carbon atom in the chain. The structure is linear with single bonds denoted by lines between the atoms. The first and last carbon atoms each have three hydrogen atoms attached. The second carbon has one hydrogen and one chlorine atom. The third carbon has two hydrogen atoms attached. This setup is critical for identifying the chiral center, which is significant in studying the compound's chemical behavior.

[Generated by AI]

The chiral carbon atom is the carbon atom bonded to the chlorine atom (marked with a red asterisk). It is bonded to a chlorine atom, a CH_3, a C_2H_5 group and a hydrogen atom.



Nature of Science

Aspect: Patterns and trends

The number of enantiomers is related to the number of chiral carbon atoms in the molecule. There are 2^n optical isomers for an organic compound with *n* chiral carbon atoms. So butan-2-ol, with its one chiral carbon atom, has $2^1 = 2$ enantiomers (**Figure 5**).

The molecule shown in **Figure 7**, 2-bromo-3-chlorobutane, has two chiral carbon atoms. How many enantiomers does it have?

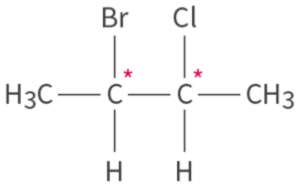


Figure 7. How many enantiomers?

Next, we will look at how to represent enantiomers using the stereochemical formula that was first introduced in section S3.2.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/types-of-formula-id-43467/). In this type of structure, the chiral carbon atom is placed at the centre and tapered bonds are used to show the orientation of the bonds. Remember that a solid tapered bond represents a bond pointing towards the viewer, a dashed tapered bond represents a bond pointing away from the viewer and solid lines represent bonds that are in the same plane as the paper (or screen) (**Figure 8**).

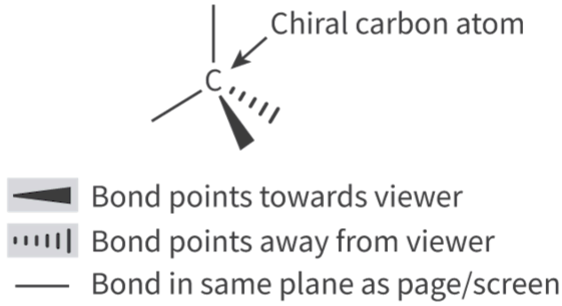


Figure 8. The arrangement of bonds in a stereochemical formula.

More information for figure 8

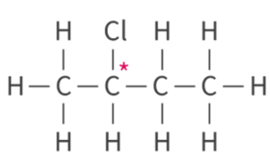
The image is a diagram illustrating a stereochemical formula, focusing on the structure around a chiral carbon atom. The central feature is the chiral carbon, labeled with a 'C'. There are three distinct types of bonds shown: 1) A solid tapered bond extending from the carbon, indicating the bond points towards the viewer. 2) A dashed tapered bond extending in another direction, showing the bond points away from the viewer. 3) A solid line bond, representing bonds that lie in the same plane as the page or screen. These visual cues help in understanding the three-dimensional orientation of the molecule in two-dimensional representation—an essential aspect of stereochemistry.

[Generated by AI]

Worked example 1

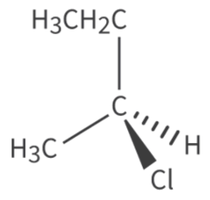
Sketch the two enantiomers of 2-chlorobutane using stereochemical formulas.

1. Start by sketching the full structural formula of 2-chlorobutane and identifying the chiral carbon atom.

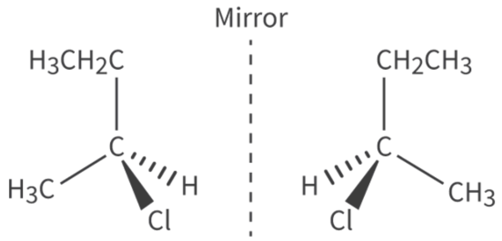


1. Place the chiral carbon atom at the centre and draw in the bonds as shown. Use lines and tapered bonds to represent the 3D structure.

1. Next add the atoms or groups of atoms to the structure.



1. Finally, draw the second enantiomer, making sure that it is a mirror image.



Use of a polarimeter to distinguish between two enantiomers

The two enantiomers can be distinguished by the difference in their interaction with plane-polarised light. Plane-polarised light is light that has been passed through a polarising filter (**Figure 9**). After passing through the polarising filter, the light oscillates in only one direction and is known as plane-polarised light.

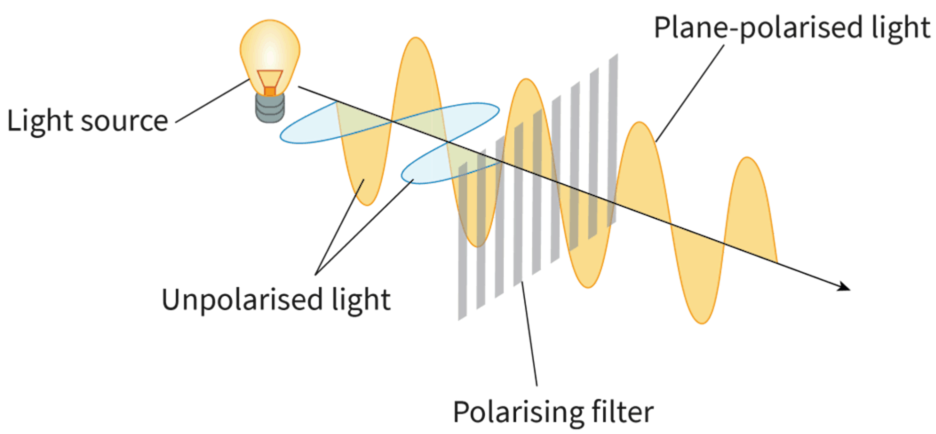


Figure 9. Plane-polarised light is produced by passing unpolarised light through a polarising filter.

More information for figure 9

The image is a diagram illustrating the process of creating plane-polarised light. On the left, a light source emits unpolarised light, represented by multiple oscillating waves in different directions. These waves encounter a polarising filter, which is depicted as a set of vertical lines. As the light passes through this filter, the oscillation of the light is restricted to a single plane, resulting in plane-polarised light. This plane-polarised light is shown as waves oscillating in a single direction beyond the filter. Labels are present to indicate the light source, unpolarised light, polarising filter, and plane-polarised light.

[Generated by AI]

The plane-polarised light is passed through a solution containing one of the enantiomers in a piece of apparatus known as a polarimeter (**Figure 10**).

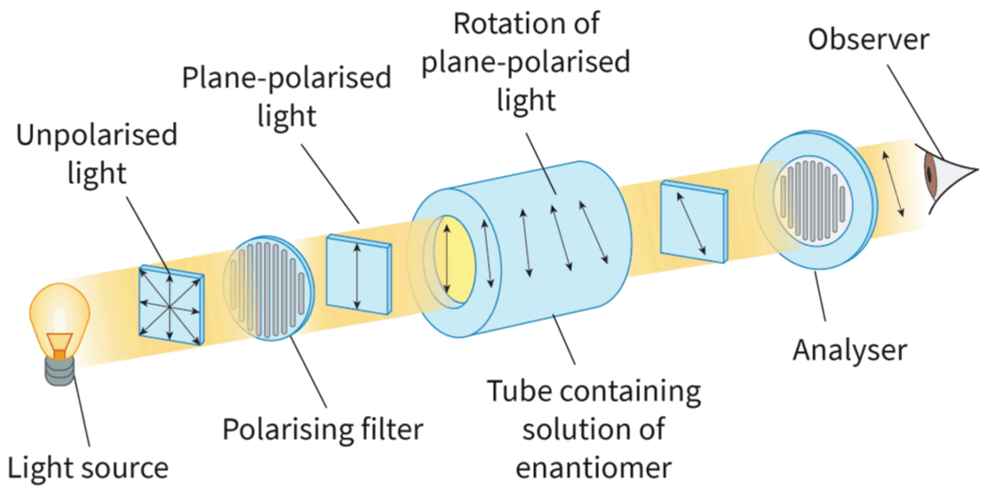


Figure 10. A polarimeter.

More information for figure 10

The image illustrates the operation of a polarimeter, showing the passage of light through its components. Starting from the left, a light source emits unpolarised light, which then passes through a polarising filter to become plane-polarised light. This plane-polarised light enters a tube containing a solution of an enantiomer, where it undergoes rotation. The rotated plane-polarised light exits the tube and passes through an analyser before reaching the observer's eye on the right. Each component (light source, polarising filter, tube, analyser, observer) is labeled accordingly within the diagram, with arrows indicating the direction of light movement.

[Generated by AI]

The operation of the polarimeter is as follows:

- Unpolarised light is passed through a polarising filter, which produces plane-polarised light.
- The plane-polarised light then passes through a solution containing the enantiomer.
- The plane-polarised light is rotated either clockwise or anti-clockwise.
- The analyser is used to measure the angle and direction of rotation of the plane-polarised light.

The two enantiomers will rotate the plane-polarised light by the same angle but in opposite directions: one will rotate the plane-polarised light clockwise and the other anti-clockwise (**Figure 11**). This is how the two enantiomers can be distinguished from each other: by the opposite rotations of the plane-polarised light.

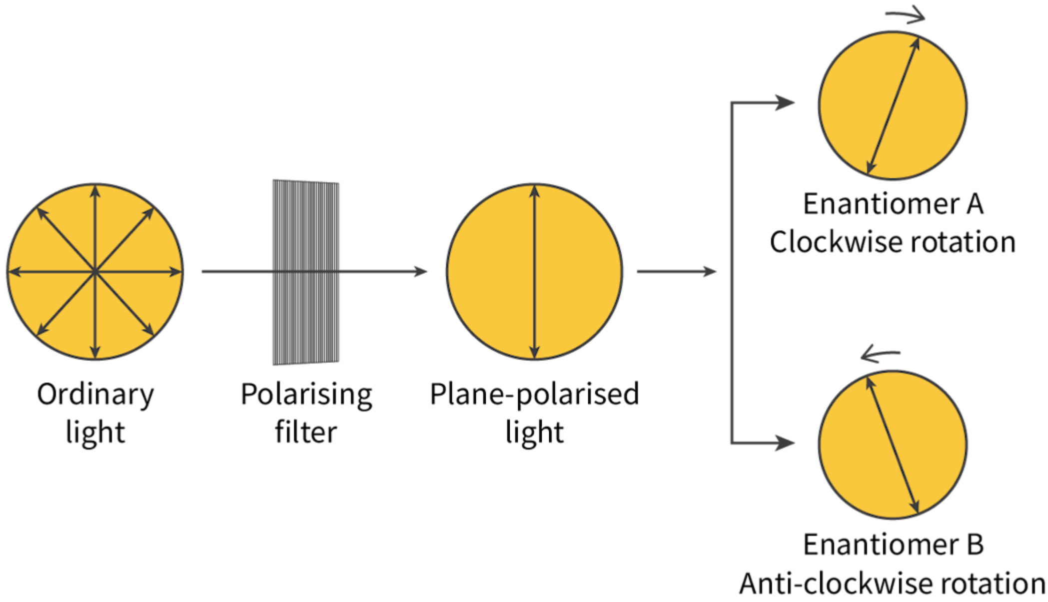


Figure 11. The two enantiomers rotate plane-polarised light by the same angle but in opposite directions.

More information for figure 11

The diagram shows how enantiomers affect the rotation of plane-polarised light. On the left, ordinary light with arrows pointing in various directions passes through a polarising filter. This filter transforms it into plane-polarised light, illustrated by arrows pointing in a single vertical direction. The plane-polarised light then encounters two enantiomers. Enantiomer A, on the top right, causes a clockwise rotation of the light, as indicated by a circular arrow around the light symbol. Enantiomer B, below it, causes an anti-clockwise rotation, highlighted by a similar but reversed arrow direction. The diagram uses arrows to indicate the path and rotation of light through each stage and labels each component clearly in text.

[Generated by AI]

So far, we have seen the effect on the rotation of plane-polarised light when the solution contains only one enantiomer. What would happen if the solution contains equal amounts of both enantiomers? Such a solution is known as a racemic mixture or a racemate. When plane-polarised light is passed through a racemic mixture, there is no net rotation, which causes the mixture to be optically inactive. **Figure 12** shows the effect of plane-polarised light passing through a racemic mixture: the mixture contains equal amounts of both enantiomers and is therefore optically inactive. Optical rotation can also be used to determine the purity of a sample containing an enantiomer: a pure sample will have its full optical rotation, but if the other enantiomer is also present as a contaminant, the optical rotation will be reduced.

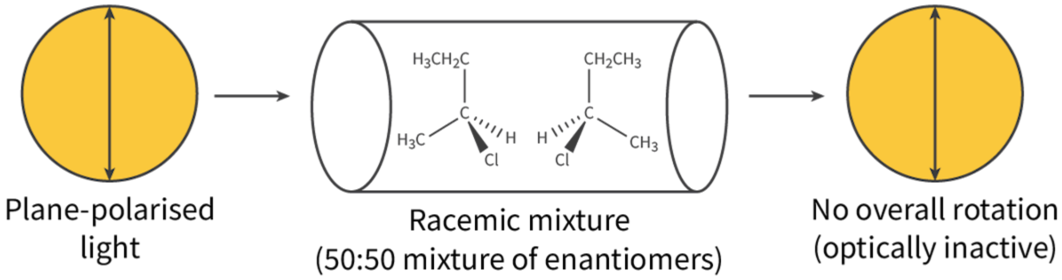


Figure 12. A racemic mixture contains equal amounts of both enantiomer and is optically inactive.

More information for figure 12

The diagram illustrates the passage of plane-polarised light through a racemic mixture. On the left, a circle represents the plane-polarised light indicated by parallel vertical arrows. In the middle, a cylindrical shape signifies the racemic mixture with equal amounts of two enantiomers, each depicted with molecular structures including CH3, H, Cl, and CH2CH3 groups. The text \'Racemic mixture (50:50 mixture of enantiomers)\' is present below the cylinder, suggesting equal concentrations of the enantiomers. On the right, another circle similar to the first one shows the resultant light, indicating \'No overall rotation (optically inactive)\'. This sequence demonstrates that a racemic mixture does not rotate plane-polarised light, rendering it optically inactive.

[Generated by AI]

Physical and chemical properties of stereoisomers

The physical properties of stereoisomers are identical apart from their interaction with plane-polarised light. They also have similar chemical properties, except for their interaction with other chiral molecules such as those found in the human body. Take for example the two enantiomers of limonene, (*R*)-limonene and (*S*)-limonene (**Figure 13**).

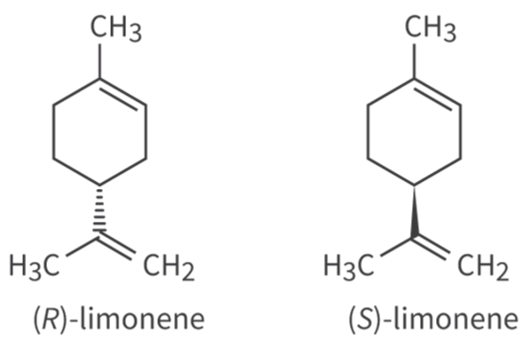


Figure 13. The two enantiomers of limonene

More information for figure 13

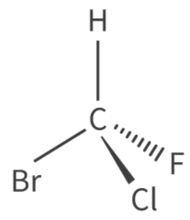
The image shows two structural diagrams of enantiomers of limonene, specifically (R)-limonene on the left and (S)-limonene on the right. Both structures feature a six-carbon ring with a double bond, denoted by the hexagonal ring structure. Attached to the ring are three additional groups: a methyl group (CH3) positioned at the top of the ring, a methylene group (CH2) at the right bottom, and a propene side chain consisting of another methyl group (CH3) and a methylene group. Both diagrams have similar structures but with different orientations at the chiral centers, indicated by wedge-shaped and dashed triangular lines, illustrating their distinct three-dimensional arrangements. These differences lead to distinct smells: (R)-limonene smells like oranges, and (S)-limonene smells like turpentine when interacting with chiral receptors in the human olfactory system.

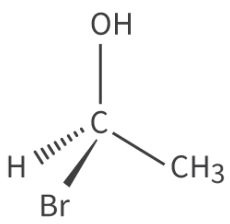
[Generated by AI]

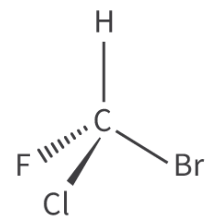
(*R*)-limonene smells like oranges and (*S*)-limonene smells like turpentine. The difference in smell is caused by the interaction of the two enantiomers with the chiral receptors in our olfactory system, the system of the human body that detects smells. Apart from their interaction with other chiral molecules, enantiomers have identical chemical properties.

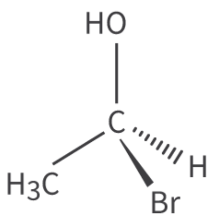
Another interesting example is the two enantiomers of glucose that are named D-glucose and L-glucose. D-glucose is the natural form of glucose that is used by the human body as an energy source. L-glucose does not occur in nature but can be produced artificially. Both enantiomers have an identical taste but L-glucose cannot be broken down by the human body so cannot be used as an energy source.

In **Interactive 3**, you will practise matching pairs of optical isomers.









Check

Interactive 3. Pair up the optical isomers.

More information for interactive 3

An interactive illustrates two chemical structures, where users need to match their corresponding optical isomer pairs by analyzing the three-dimensional arrangement of atoms around a chiral carbon.
Structure 1: On the Top Left
A central carbon atom is bonded to a hydrogen atom at the top via a solid line, a fluorine atom at the bottom right via a dashed line, a chlorine atom at the bottom via a wedge line, and a bromine atom at the bottom left via a solid line.
Structure 2: On the Bottom Left
A central carbon atom is bonded to a hydroxy group at the top via a solid line, a hydrogen atom at the bottom right via a dashed line, a bromine atom at the bottom via a wedge line, and a methyl group at the bottom left via a solid line.
Both the structures have drop boxes on the right.
Drag-and-drop structures on the right are described as follows:
Option 1: On the Top Right
A central carbon atom is bonded to a hydroxy group at the top via a solid line, a methyl group at the bottom right via a solid line, a bromine atom at the bottom via a wedge line, and a hydrogen atom at the bottom left via a dashed line.

Option 2: On the Bottom Right
A central carbon atom is bonded to a hydroxy group at the top via a solid line, a bromine atom at the bottom right via a solid line, a chlorine atom at the bottom via a wedge line, and a fluorine atom at the bottom left via a dashed line.
Read below for the solution:
The correct pairs of optical isomers are:
Structure 1 corresponds to Option 2
Structure 2 corresponds to Option 1

Try this activity to test your understanding of optical isomers.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual activity

1. Determine if the following molecules contain a chiral carbon atom. Use molview (https://molview.org/) to view the molecules in 3D. Type in the IUPAC name of the compound, shown in brackets.

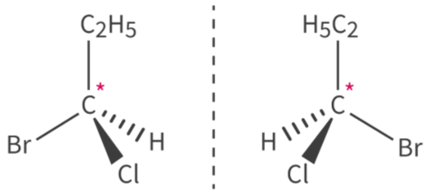
   (a) $CH_3CH_2CHBrCH_2CH_3$ (3-bromopentane)
   (b) $CH_3CClBrCH_3$ (2-bromo-2-chloropropane)
   (c) $CH_3CH_2CHBrCl$ (1-bromo-1-chloropropane)
   (d) $CH_3CHClCH_2CH_2Br$ (1-bromo-3-chlorobutane)

2. For each molecule that contains a chiral carbon atom, sketch the stereochemical formula of both enantiomers (mirror images).

   1. (a) 3-bromopentane does not have a chiral carbon atom. The carbon atom bonded to the bromine is bonded to two C_2H_5 groups and a H atom that makes only three different groups.
      (b) 2-Bromo-2-chloropropane does not have a chiral carbon atom. The carbon atom is bonded to the bromine atom and the chlorine atom is bonded to two CH_3 groups that only makes three different groups.
      (c) 1-bromo-1-chloropropane does have a chiral carbon atom. The carbon atom bonded to the Br and Cl atoms is bonded to four different atoms or groups of atoms: a Cl atom, a Br atom,

a C_2H_5 group and a hydrogen atom. This makes four different groups.
(d) 1-bromo-3-chlorobutane does have a chiral carbon atom. The third carbon atom is bonded to four different atoms or groups of atoms: a CH_3 group, a Cl atom, a C_2H_4Br group and a hydrogen atom. This makes four different groups.

2. Two enantiomers of 1-bromo-1-chloropropane



1. Two enantiomers of 1-bromo-3-chlorobutane